

Winfried Ripken

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Google scholar
GitHub
LinkedIn

SKILLS

ML tools:

Jax • PyTorch • Numpy • SciPy •
Pandas • OpenCV

MLOps:

Docker • Kubernetes • Terraform •
GCP • Weights & Biases • Slurm

Coding:

Python (proficient) • C++ (working
knowledge) • C#

Molecular Simulation:

E3x • Atomic Simulation Environment
(ASE) • SchNetPack • RDKit

Languages:

German • English • French (basic)

COURSEWORK

GRADUATE

Machine Learning on Graphs
Probabilistic Graphical Models
Big Data and Visualization
Predictive Analytics
NLP using word vectors

UNDERGRADUATE

Linear Algebra
Probability Theory
ML & Computer Vision
Physics

EXPERIENCE

TECHNICAL UNIVERSITY BERLIN | PHD STUDENT

Apr 2024 - present | Berlin, Germany

I am a 3rd-year PhD student supervised by Stefan Chmiela and Klaus-Robert Müller.

- My most recent project enables Molecular Dynamics (MD) simulations with variable time steps, at least an order of magnitude larger than typical integration steps. The operator is trained without trajectory supervision using a tailored consistency loss and independent pairs of positions and forces.
- I co-developed an efficient generative modeling architecture for molecular modeling based on (equivariant) diffusion transformers (DiTs). Using flow matching, we reach SOTA results on several molecular conformer generation benchmarks, significantly shifting the speed-accuracy pareto front.
- I co-developed a novel framework for diffusion-based generative modeling for molecules and images. It disentangles signal to noise ratio (SNR) from total variance (TV) to enable more efficient diffusion schedules, reaching high quality results with as few as 4 sampling step. Currently in submission.

MERANTIX MOMENTUM GMBH | MACHINE LEARNING RESEARCHER

Nov 2021 - Mar 2024 | AI Campus Berlin, Germany

Merantix Momentum conducts ML research for real world applications.

- I was the lead in two research projects, where we collaborated with Volkswagen to improve electric motor design via automated simulation of PDEs via surrogate models, and with other German manufacturing players to introduce ML-based decisions into their assembly lines.
- Trained a solution operator for Partial Differential Equations (PDEs) using Graph Neural Networks (GNNs): I developed a novel augmentation technique that improved generalization performance by more than one order of magnitude, as well as a graph rewiring strategy enabling training solution operators for very high-resolution data (more than 30K nodes).

TERAKI GMBH | VIDEO DEEP LEARNING INTERN & WORKING STUDENT

Jul 2020 - Mar 2021 | Berlin, Germany

Teraki provides software for teleoperated and autonomous driving.

- Trained a very lightweight yet accurate semantic segmentation network for road scenes (60 FPS on Jetson Nano, false negative rate < 2%).
- Pruned 50% of the weights to increase FPS by another 30%.
- I contributed to client projects & presented my results in client calls.

HASSO-PLATTNER-INSTITUTE | TEACHING (DESIGN THINKING)

Sep 2019 - Jun 2020 | Potsdam, Germany

I supervised 2 student ML projects, resulting in one filed patent.

JOYSON SAFETY SYSTEMS | SOFTWARE ENG. WORKING STUDENT

Aug 2014 - Jul 2017 | Elterlein, Germany

EDUCATION

TECHNICAL UNIVERSITY BERLIN | PHD IN COMPUTER SCIENCE

Apr 2024 - present | Berlin, Germany

HASSO PLATTNER INSTITUTE MSc IN COMPUTER SCIENCE

Oct 2017-Nov 2021 | Potsdam, Germany | Grade: 1.1 (1.0 = best, 4.0 = pass)

Exchange program (ME310) with Stanford university from Oct 2017 until Jun 2018.

Master project: Self-supervised learning for 3D data (Published at NeurIPS 2020).

Master thesis: User-controllable Neural Style Transfer (Published at ECCV 2022).

EFREI PARIS SUD ERASMUS+ SEMESTER

Aug 2018-Feb 2019 | Villejuif, France

Best grade for ML research project with Orange in Paris.

CHEMNITZ UNIVERSITY BSc IN COMPUTER SCIENCE

Oct 2014-Sep 2017 | Chemnitz, Germany | Grade: 1.1 (1.0 = best, 4.0 = pass)

Bachelor thesis about asynchronous Deep Reinforcement Learning (best grade 1.0).

SELECTED PUBLICATIONS

Ripken, Winfried*, Michael Plainer*, Gregor Lied*, Thorben Frank*, Oliver T. Unke, Stefan Chmiela, Frank Noé, and Klaus-Robert Müller (2026). "Learning Hamiltonian Flow Maps: Mean Flow Consistency for Large-Timestep Molecular Dynamics". In: *ICML (Spotlight, top 2.2%)*.

Frank, J. Thorben*, **Winfried Ripken***, Gregor Lied*, Klaus-Robert Müller, Oliver T. Unke, and Stefan Chmiela (2025). "Sampling 3D Molecular Conformers with Diffusion Transformers". In: *NeurIPS*.

Ohler, Simon, Daniel Brady, Patrick Mischke, Jana Bender, Herwig Ott, Thomas Niederprüm, **Winfried Ripken**, Johannes S. Otterbach, and Michael Fleischhauer (2025). "Nonequilibrium universality of Rydberg-excitation spreading on a dynamic network". In: *Phys. Rev. Res.*

Siems, Julien*, Konstantin Ditschuneit*, **Winfried Ripken***, Alma Lindborg*, Maximilian Schambach, Johannes S Otterbach, and Martin Genzel (2023). "Curve Your Enthusiasm: Concurvity Regularization in Differentiable Generalized Additive Models". In: *NeurIPS*.

Lötzsch, Winfried*, Max Reimann*, Martin Büsselmeier, Amir Semmo, Jürgen Döllner, and Matthias Trapp (2022). "WISE: Whitebox Image Stylization by Example-Based Learning". In: *ECCV*.

Taleb, Aiham, **Winfried Löttsch**, Noel Danz, Julius Severin, Thomas Gaertner, Benjamin Bergner, and Christoph Lippert (2020). "3D Self-Supervised Methods for Medical Imaging". In: *NeurIPS*.

PREPRINTS

Maeß, Johannes, Leon Werner, J Thorben Frank, **Winfried Ripken**, Martin Michajlow, Joshua Futterer, Klaus-Robert Müller, and Stefan Chmiela (2026). "Implicit Machine Learning Force Fields Accelerate Molecular Dynamics Simulations". In: *arXiv preprint arXiv:2607.29158*.

Kahouli, Khaled, **Winfried Ripken**, Stefan Gugler, Oliver T Unke, Klaus-Robert Müller, and Shinichi Nakajima (2025). "Enhancing Diffusion Models Efficiency by Disentangling Total-Variance and Signal-to-Noise Ratio". In: *arXiv preprint arXiv:2502.08598*.

* indicates equal contribution

See **scholar** for the full list of publications.